Switching from Sphinx to Markdown

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PyCon AU 2026





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Acknowledgement of Country

I am from the lands of the
Bodéwadmi and **Odaawaa** people.

We are on the lands of the **Turrbal** people.

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reStructuredText (reST) and Sphinx

- What is it?
- History
- Documentation generation
- Ecosystem inertia

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Markdown

- What is it?
- History
- Adoption
- AI integration
- Documentation options

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Pros and cons of reST

- What it does well
- Difficulties it presents

```
.. role:: strike
    :class: strike

================
Document Heading
================

A paragraph with **bold
text**, :strike:`strikethrough text`,
and a `link to BeeWare <https://
beeware.org>`_.
```




Pros and cons of Markdown

- What it does well
- Difficulties it presents

```
# Document Heading

A paragraph with **bold text**,
~strikethrough text~, and a
[link to BeeWare](https://
beeware.org).
```




Which one is better?

It depends.



Migrating BeeWare: Origin story

- How it began
- Becoming a contributor

> *"Once upon a time, reST seemed like a good idea, but I'm increasingly thinking moving to Markdown across the board would be worth it."*
>
> *"If only we had a sprint coming up..."*



Migrating BeeWare: The full scope

- Four categories:
  - Tutorial
  - Major tools: Toga, Briefcase, Rubicon Obj-C
  - Standalone documents
  - Website



Website content

- Lektor in reST mode
- Pandoc and Lektor API
- Proof of reST to Markdown

A few hours...

+17,284 -8,448



Preparation for project migration

- Consider options
- Investigate MkDocs
- Translations
- Explore add-ons
- Migration automation
- Tools

One and a half months...



Preparation: Consider options

- Requirements
- Sphinx and MyST parser
- MkDocs
- Other alternatives



Preparation: Investigate MkDocs

- Learn the basics
- Research features
- Material theme
- Establish viability



Preparation: Build and translations

- No viable translation options
- Translate Toolkit
- Build tooling
- Build integration



Preparation: Explore add-ons

- `attr_list`
- `mkdocs-autorefs`
- `mkdocs-macros`
- `mkdocs-literate-nav`
- `mkdocstrings`
- PyMDown Extensions (`pymdownx`)



Preparation: Customise Pandoc

- Pandoc Markdown extensions
- CommonMarkX
- Tailoring the Pandoc command

```
find -f **/*.rst -type f -exec sh -c pandoc -f rst "${0}"
    -t commonmark_x-bracketed_spans-smart-alerts
    -o "$(dirname {})/$(basename {} .rst).md" {} \;
```




Preparation: Conversion scripts

- Content
- Docstrings
- Translations

```
r"((#*)?( )?`(~)?([A-Za-z0-9\n. \-_!'\"]*)(?:\(\))?
(?:<)?(?:\/)?([.a-z0-9\/\-]*)?>?`{.interpreted-
text(?:\n)?(?: )?role=\"([a-z]+)\"})"
```




Preparation: Linting and checks

- rumdl
- PySpelling
- markdown-checker



Preparation: BeeWare Docs Tools

- Sandbox for feature testing
- Single source of dependencies
- Theme unification
- Shared content
- Build tooling



Tutorial

- Good first project
- Relevant basic features
- Translated content

Two weeks...

+86,891 -72,072



Toga

- Most complex project
- Proof of concept
- Configuration
- Additional changes
- Manual verification

Two and a half months...

+9,990 -11,721

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Briefcase

- Less complicated
- Utilised existing tooling

Three days...

+7,524 -9,520



Rubicon Objective-C

- Typing file to the rescue
- Unusual complications

Two weeks...

+4,728 -6,077



Website

- Stretched the capabilities
- Simplified content creation
- Unified tooling

Three months...

+244,319 -174,505



Standalone documents

- Change note system
- Templates
- README



The human element

- Presenting a successful proposal
- Deliverable, not just doable
- Stages
- Engaging the community



Total time:

Nine+ months

37 PRs · 2,823 files · +390,608 −306,950

697,558 lines touched · Net +83,658

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Was it worth it?

Absolutely.

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Would we do it again?

Unequivocally: Yes.

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Should you do it?

You decide.



The future of MkDocs

- MkDocs 1.x: not being maintained
- MkDocs 2.0: not backwards compatible
- ProperDocs: fork to preserve 1.x
- Zensical: ground-up rebuild compatible with 1.x
- General migration process is the same



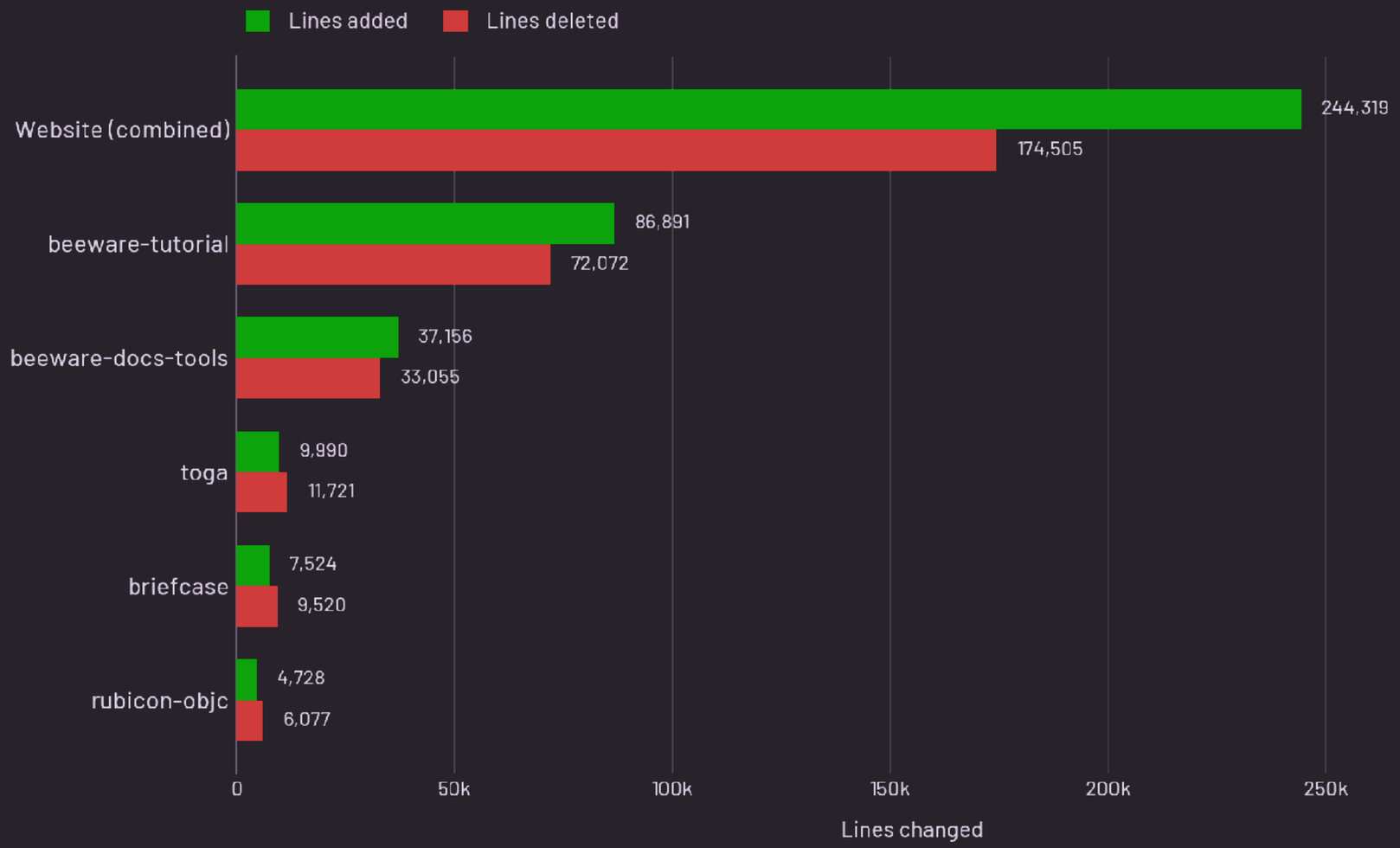
Find me:



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Slides and resources: https://kattni.com/PyConAU2026

Lines touched: by repo



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Breakdown of changes

Group	PRs	Files	Added	Deleted	Churn	Net
Website (combined)	11	1,284	244,319	174,505	418,824	69,814
beeware-tutorial	9	396	86,891	72,072	158,963	14,819
beeware-docs-tools	7	187	37,156	33,055	70,211	4,101
toga	4	574	9,990	11,721	21,711	-1,731
briefcase	3	228	7,524	9,520	17,044	-1,996
rubicon-objc	3	154	4,728	6,077	10,805	-1,349

